Mid Semester Exam

Numerical Analysis (MA5060 and MA3060)

Full Marks: 31
Time- 1 hour 30 minutes

Solve the problem given below:

1. For a given IVP $y' = f(x, y),\ \ y(x_0) = y_0,\ x \in [x_0, b]$, apply a numerical method

$$y_{n+1} = y_n + \frac{h}{2}\left[f(x_n, y_n) + f\left(x_{n+1}, y_n + hf(x_n, y_n)\right)\right], \text{ for } n = 0,1,2\ldots, N,$$

where $f(x, y) = 1 + a(1 + x - y),\ a > 0,\ x_0 = 0,\ y_0 = 2,\ b = 1,\ y(x_{n+1}) \approx y_{n+1}$, $y(x_n) \approx y_n\ , x_n = x_0 + nh,\ n = 1, 2, \ldots, N,\ h$ is the step length and N is the number of equally spaced subintervals in $[x_0, b]$. Explain the geometrical interpretation of this numerical method. By applying this method, find the local truncation and global errors, and their bounds. Is this method consistent and what is the order of the method? Justify. Discuss the convergence of this method. Now solve the IVP by the given method and find $y(0.2)$ with the step length $h = 0.1$.

$[1 + 6 + 2 + 3 + 3 = 15$ marks$]$

2. Apply Newton's method to the function $f(x) = \begin{cases} x^{\frac{2}{3}}, & x \geq 0\,, \\ -x^{\frac{2}{3}}, & x < 0\,, \end{cases}$ with the root $\alpha = 0$.

What is the behavior of the iterates? Do they converge, and if so, at what rate? Justify your answers. [5 marks]

3. The iteration $x_{n+1} = 2 - (1 + c)\, x_n + c\, x_n^3,\ n \geq 0$ will converge to $\alpha = 1$ for some values of c (provided x_0 is chosen sufficiently close to α). Find the values of c for which this is true. For what value of c will the convergence be quadratic? Justify your answers. [5 marks]

4. Given that α is a fixed positive real number, consider the function g defined on the interval

$[0,1]$ by
$$g(x) = \begin{cases} 2^{-\left[1 + \left\{\log_2\left(\frac{1}{x}\right)\right\}^{\frac{1}{\alpha}}\right]^{\alpha}}, & x \in (0,1] \\ 0, & x = 0 \end{cases}$$

Then find the fixed point of $g(x)$ in the interval $[0,1]$. Consider the sequence $\{x_k\}$ defined by the iteration $x_{k+1} = g(x_k),\ k \geq 0$ with $x_0 = 1$. Is the fixed-point stable by justifying the convergence of the sequence $\{x_k\}$? Discuss the speed of the convergence depending on α .

[6 marks]

***********************Best of Luck***********************